

BASICS OF ECONOMICS

MGT404

Production Functions and Costs

Yale

Recap and Today's Plan

Last Lecture

- Government interventions
- Calculating the tax burden and deadweight loss—how do we do it?
 - The inelastic side of a market carries more of the burden (incidence) of a tax
- Externalities and corrective taxation

Today—Firms

- Types of Costs
- The Production Function
- Cost Minimization
 - Finding “Marginal Product”
 - Equating MPI to find optimal production function
- Firm supply function = firm's marginal cost function at optimal inputs
- Market supply function is aggregate of firm supply functions.

Today's Plan: producing optimally



1. Types of Costs

- **Fixed versus variable costs**
- ***Marginal* cost**—the cost of making one more

2. The Production Function

- Variable costs don't just come from the sky—they're based on what inputs are required

3. Cost Minimization

- Need to find the best mix of inputs
- Want to know it for every possible quantity
- Use the *marginal product of investing* in inputs to find the optimal production mix

4. Discussion

- In the long run, can we do better?
- Re-optimizing production decisions

Introduction: How does a warehouse look like?



Or



Article Discussion: Capital – Labor tradeoffs (Part 1)

Question: What kinds of costs does Amazon face for logistics?

- Warehouse space
- Machines
- Workers
- Managers
- Packing supplies
- Shelves
-



Some of these scale with quantity, and some do not.

Some of these are more discretionary than others.

Some of these costs could be recovered later (i.e. selling used equipment),
some cannot.

Fixed Costs vs Variable Costs

Fixed Cost

Cost that must be incurred before any output can be produced

Examples

- Rent
- Utilities
- “Overhead” (billing department)
- Licenses, taxes, fees for production

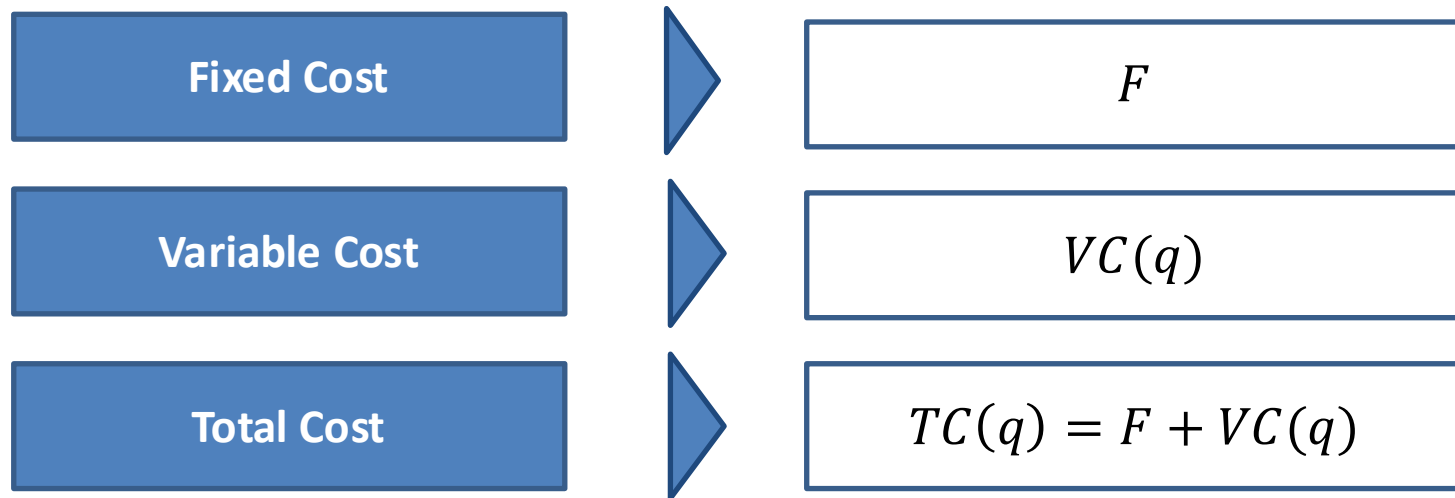
Variable Cost

Cost that scales with the amount of output produced

Examples

- Raw materials
- Labor (assembly)
- Patents (per unit)
- Shipping (per unit)

Functions to Represent Costs



Example: If Total Cost is given by

$$TC(Q) = \underbrace{10}_{\text{Fixed Cost}} + \underbrace{12Q + 0.4Q^2}_{\text{Variable Cost}}$$

Average Cost and Marginal Cost

Average cost

Total cost on a per-unit basis

$$\begin{aligned} AC(q) &= \frac{TC(q)}{q} \\ &= \frac{F}{q} + \frac{VC(q)}{q} \end{aligned}$$

Q: Should I be paying the fixed cost to produce?

A: If price is higher than AC

“Extensive Margin”

Marginal cost

The cost of producing one additional unit of output

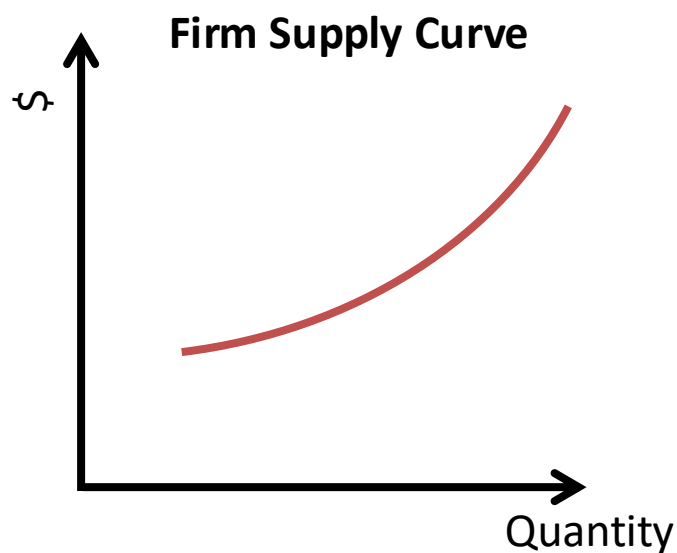
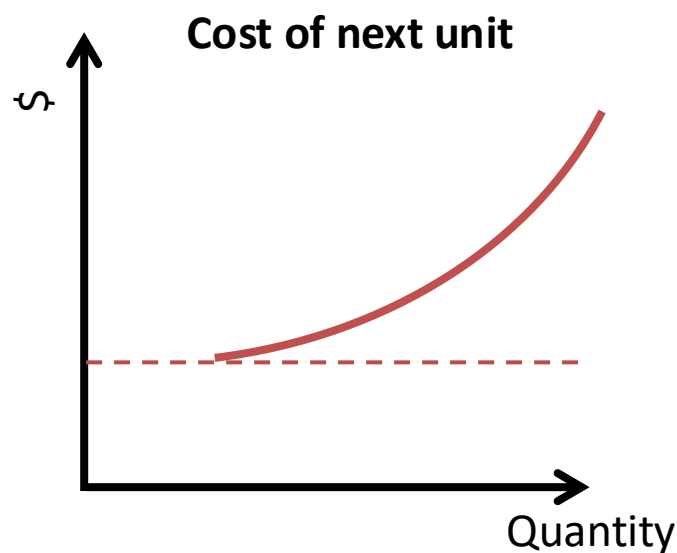
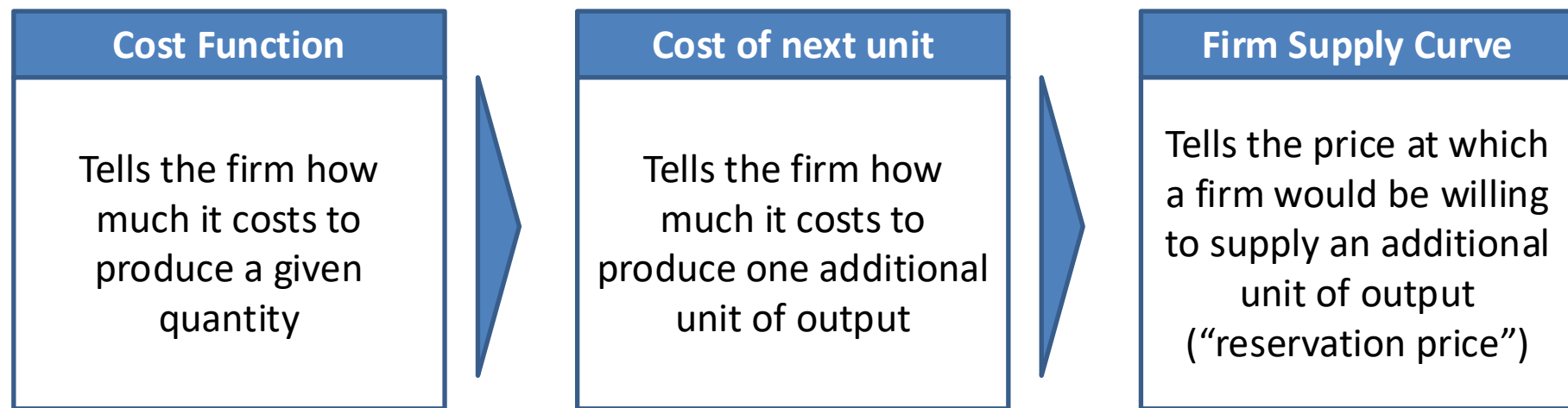
$$\begin{aligned} MC(Q) &= \frac{TC(q + \Delta q) - TC(q)}{\Delta q} \\ &= TC'(q) \\ &= VC'(q) \end{aligned}$$

Q: Should I make one more unit?

A: If price is higher than MC

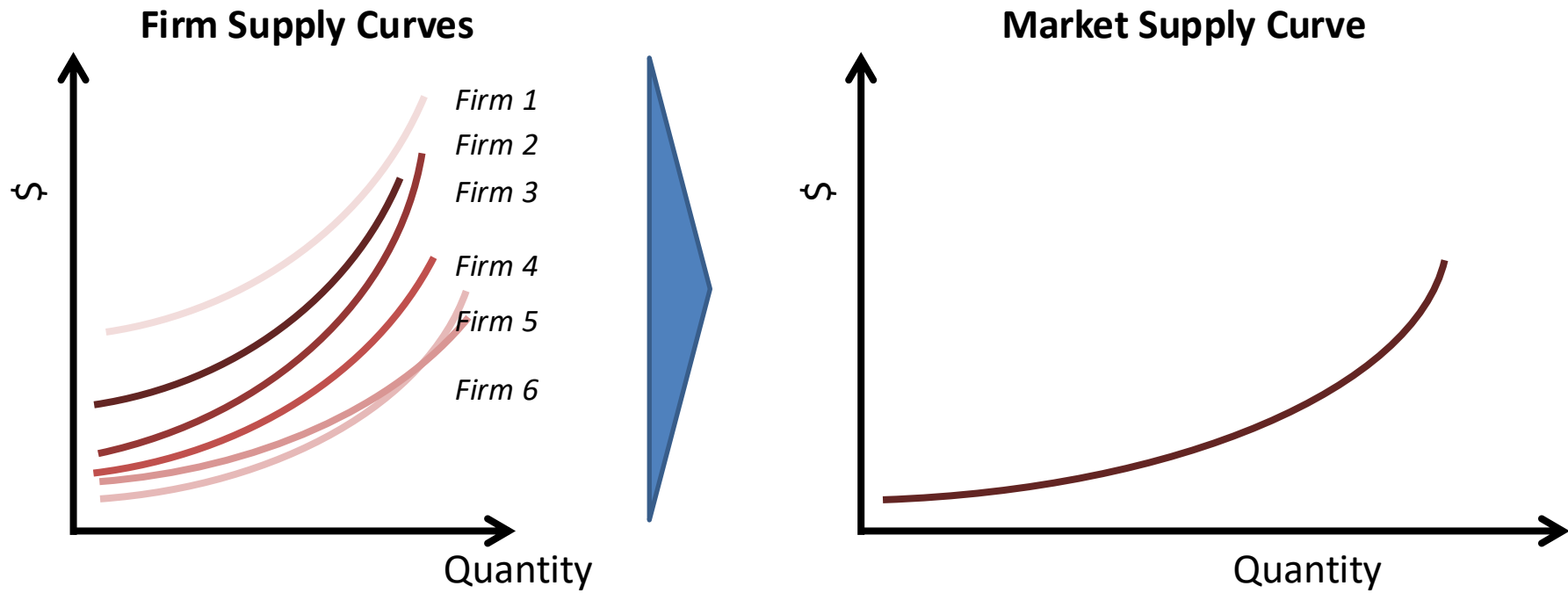
“Intensive Margin”

How do firms choose supply?



Firm marginal cost curve = Firm supply curve

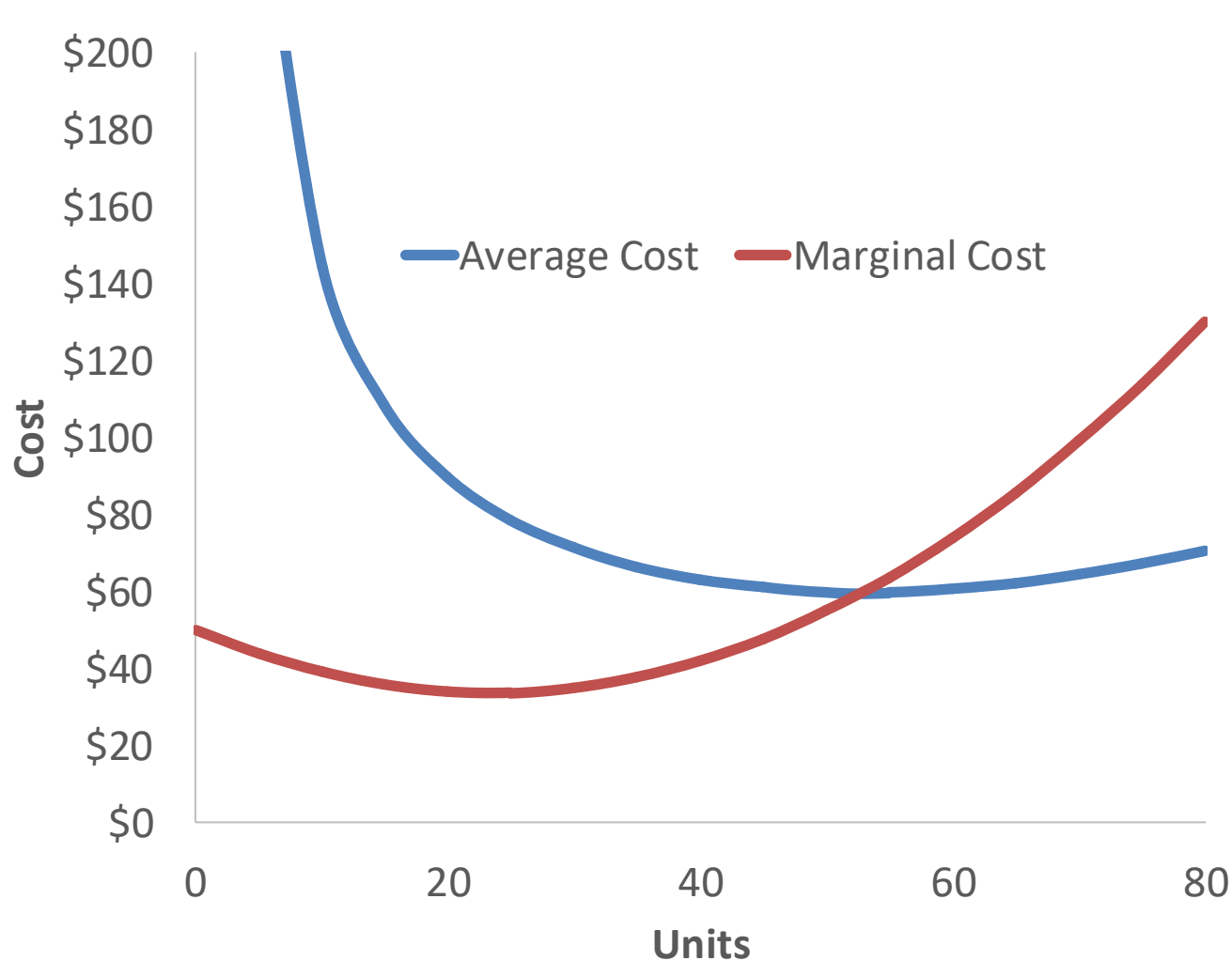
Combination of All Firms



Market Supply Curve

As with demand, we can aggregate individual firm supply curves into the market supply curve for a good. Here there are multiple firms with upward-sloping cost curves at different levels, allowing us to find a price-quantity relationship for the market as a whole.

Graphing cost curves



- The two curves are clearly related:
- When $MC < AC$, what is AC doing?
 - *Falling*
- When $MC > AC$, what is AC doing?
 - *Rising*
- So, what can we say about where the two cross?
 - They cross at the $\min(AC)$.

Summary of Costs

Fixed Costs (FC)

Cost that must be incurred before any output can be produced

- E.g. rent, utilities, licenses

versus

Variable Costs (VC)

Cost that scales with the amount of output produced

- E.g. labor, raw materials

Average Cost (AC)

Total cost on a per-unit basis
Can be ATC or AVC

versus

Marginal Cost (MC)

The cost of producing one additional unit of output

- Should I be in this business at all? If price > average cost, yes.

- Should I increase output? If price > marginal cost, yes.

The Other Cost you Hear Economists Talk About: Opportunity Cost

The cost of the next best alternative that your choice precludes. E.g. hitting “snooze” when your alarm goes off means you won’t have time for breakfast; or, when a firm pays out profits as dividends, it can no longer invest that money into more R&D.

Key Difference: Short Run vs Long Run Costs

Cost	Horizon to adjust
Electricity usage	• Today (I can turn off machines)
Raw materials	• One week (next scheduled delivery)
Labor	• Next contract negotiation (union)
Factory space	• Lease expiration (years?)



Clearly some costs are more “current”. Short enough horizon: almost all are fixed. Long enough horizon: almost all are variable.
Longer time horizon → easier to adjust → more elastic

Today's Plan: producing optimally

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2. The Production Function

- **Costs don't just come from the sky—they're based on what inputs are required**

3. Cost Minimization

- Need to find the best mix of inputs
- Want to know it for every possible quantity
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4. Discussion

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- Re-optimizing production decisions

Production Function

Production Function

If there are n inputs to production, then we summarize our output capabilities with a **production function** of n variables:

$$q = f(\text{input}_1, \text{input}_2, \text{input}_3, \dots)$$

For our Amazon Shipping example, suppose we know that:

$$q = f(l, m) = 2l + 5m + 0.5 \cdot l \cdot m$$

workers (labor)

machines

Output (in M of packages)

A production function gives us a mapping from inputs to outputs.

Production Function

Suppose we want to ship a total of 60M packages. We can use the production function to come up with multiple production plans that would achieve that.

$$q = f(l, m) = 2l + 5m + 0.5 \cdot l \cdot m$$

Plan 1

- 30 workers ($l=30, m=0$)
 $q = f(30, 0) = 2 \cdot 30 = 60$

Plan 2

- 22 workers + 1 machine ($l=22, m=1$)
 $q = f(22, 1) = 2 \cdot 22 + 5 \cdot 1 + 0.5 \cdot 22 \cdot 1 = 60$

Plan 3

- 12 machines ($l=0, m=12$)
 $q = f(0, 12) = 5 \cdot 12 = 60$



All three plans would get us to our target.

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Production Optimization – Know the Logic

Production Function

- Maps inputs into outputs.



Step 1: given costs of various inputs, solve for an optimal ratio of inputs by equating the marginal productivity of investing in each.

Optimal Input Mix

- Tells us the lowest-cost way to combine inputs *at current input prices*.

Math Not Tested Beyond Optimal Input Mix



Step 2: Plug optimal mix into production function. Solve for an optimized production function at current costs, and optimal inputs .

Optimal Production Function

- Tells us the lowest possible cost way of producing a given level of output *at current input prices*.



Step 3: Create a cost function using input prices and optimal input levels. Take the derivative to get marginal cost, which is a firm's supply curve.

Firm Supply Curve

- Tells us how much the firm will supply at different market prices.

Step 1: Marginal Productivity

Marginal Productivity

For a given input, we can calculate its **marginal productivity** – the extra output that can be produced per additional amount of that input added to the production process (here, a *partial derivative*)

$$MP_{\text{input}_1}(\text{input}_1, \text{input}_2) = \frac{\partial f(\text{input}_1, \text{input}_2)}{\partial \text{input}_1}$$

For Amazon:

Production function

$$f(l, m) = 2l + 5m + \frac{1}{2} l \cdot m$$

$$MP_l(l, m) = \frac{\partial f}{\partial l} = 2 + \frac{1}{2} m$$

$$MP_m(l, m) = \frac{\partial f}{\partial m} = 5 + \frac{1}{2} l$$

The marginal productivity is a partial derivative (i.e. treats all other variables as constants). It changes with the level of inputs you are using.

*Partial derivative
with respect to l*

*Partial derivative
with respect to m*

Step 1: Marginal Productivity of Investment

Marginal Productivity
(MP_x)

- Tells us the increase in output from using more of an input.
- **Problem: inputs often have different costs.**

Marginal Productivity of
Investment (MPI_x)

- Tells us the increase in output from spending an extra dollar on an input (i.e. bang-for-the-buck)

$$MPI_{input_1} = \frac{\Delta output}{\Delta cost} = \frac{MP_{input_1}}{p_{input_1}} = \frac{\frac{\partial f(input_1, input_2)}{\partial input_1}}{p_{input_1}}$$


- Now we have a way of comparing inputs on a comparable basis.
- Remember: MPI is just MP divided by an input's unit cost

Step 1: Marginal Productivity of Investment

We need costs of the inputs

$p_l = \$60\text{K}$
 $p_m = \$150\text{K}$

Recall MPs

$MP_l = 2 + \frac{1}{2}m$ $MP_m = 5 + \frac{1}{2}l$

Calculating MPI

$$MPI_l = \frac{\frac{\partial f(l,m)}{\partial l}}{\$60\text{K}} = \frac{MP_l}{\$60\text{K}} = \frac{2 + \frac{1}{2}m}{\$60\text{K}}$$
$$MPI_m = \frac{\frac{\partial f(l,m)}{\partial m}}{\$150\text{K}} = \frac{MP_m}{\$150\text{K}} = \frac{5 + \frac{1}{2}l}{\$150\text{K}}$$

Labor()	Machines()	Cost	MPI_l (pcks/\$1K)	MPI_m (pcks/\$1K)	Improvement?
30	0	\$1.8M	0.033	0.133	Fire l , rent m
22	1	\$1.47M	0.042	0.107	Fire l , rent m
0	12	\$1.8M	0.133	0.033	Fewer m , hire l

Is there another combination of (l, m) that can do better?

Step 1: Optimal Ratio of Inputs

Cost Minimization

Costs are minimized only if the MPIs are equal.

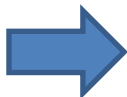
Firms always allocate funds to the inputs that give them the most return.

Optimal Production

Using the cost minimization rule, we can solve for the “optimal” production function that always selects the best ratio of inputs. **This would change if input prices change.**

In our example:

$$\frac{2 + \frac{1}{2}m}{\$60K} = \frac{5 + \frac{1}{2}l}{\$150K}$$



Solving, we get

$$l^* = 2.5m^*$$

The optimal mix has 2.5 workers per robot.

No math tested
beyond this

Step 2: Optimal Production

Math Not Tested

Optimal Production Ratio

Our optimal ratio:

$$\frac{2 + \frac{1}{2}m}{\$60K} = \frac{5 + \frac{1}{2}l}{\$150K}$$

Solving, we get

$$l^* = 2.5m^*$$

At these input prices, the optimal mix has 2.5 workers per robot.

Then: Plug in for Optimal Production Function

Plug into our production function to get cost-minimizing production function:

Sub in $l^ = 2.5m^*$*

$$f(2.5m^*, m^*) = 10m^* + 1.25m^{*2}$$

$$q = 10m^* + 1.25m^{*2}$$

Optimal Production Function

Our optimized production function is now just in terms of machines; however, it is assuming you're actually going to use the optimal amount of labor per machine.

Step 2: Finding required input amounts

The Concept

Producing using the cost-minimizing ratio, we can find the inputs required for any desired level of output.

Example

If Amazon wants to ship 60m packages, we have $q = 60$ on the left, and solve for how many “m” machines (robots) they should rent:

$$\begin{aligned}q &= 10m^* + 1.25m^{*2} \\60 &= 10m^* + 1.25m^{*2} \\1.25m^{*2} + 10m^* - 60 &= 0 \\m^* &= \frac{-10 \pm \sqrt{10^2 - 4(1.25)(-60)}}{2 * 1.25} = \frac{-10 \pm 20}{2.5}\end{aligned}$$

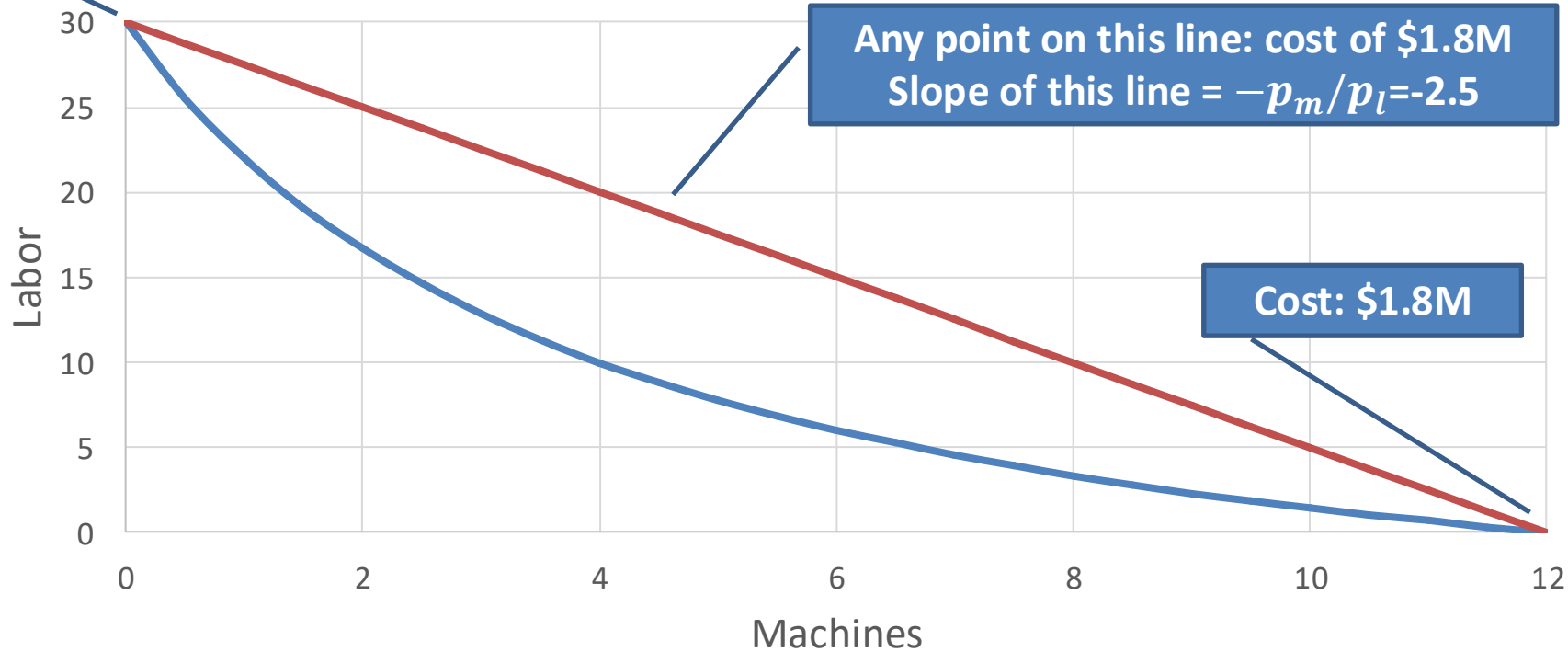
- Optimal input levels for $q = 60$ are $m^* = 4 \rightarrow l^* = 10$
- Total cost: $4 * 150K + 10 * 60K = \$1.2M$
- What about for different levels of output? We can do this same exercise but instead solving for a general q .

Graphically, What Are We Doing?

Cost: \$1.8M

Production Plans that Achieve 60M Output

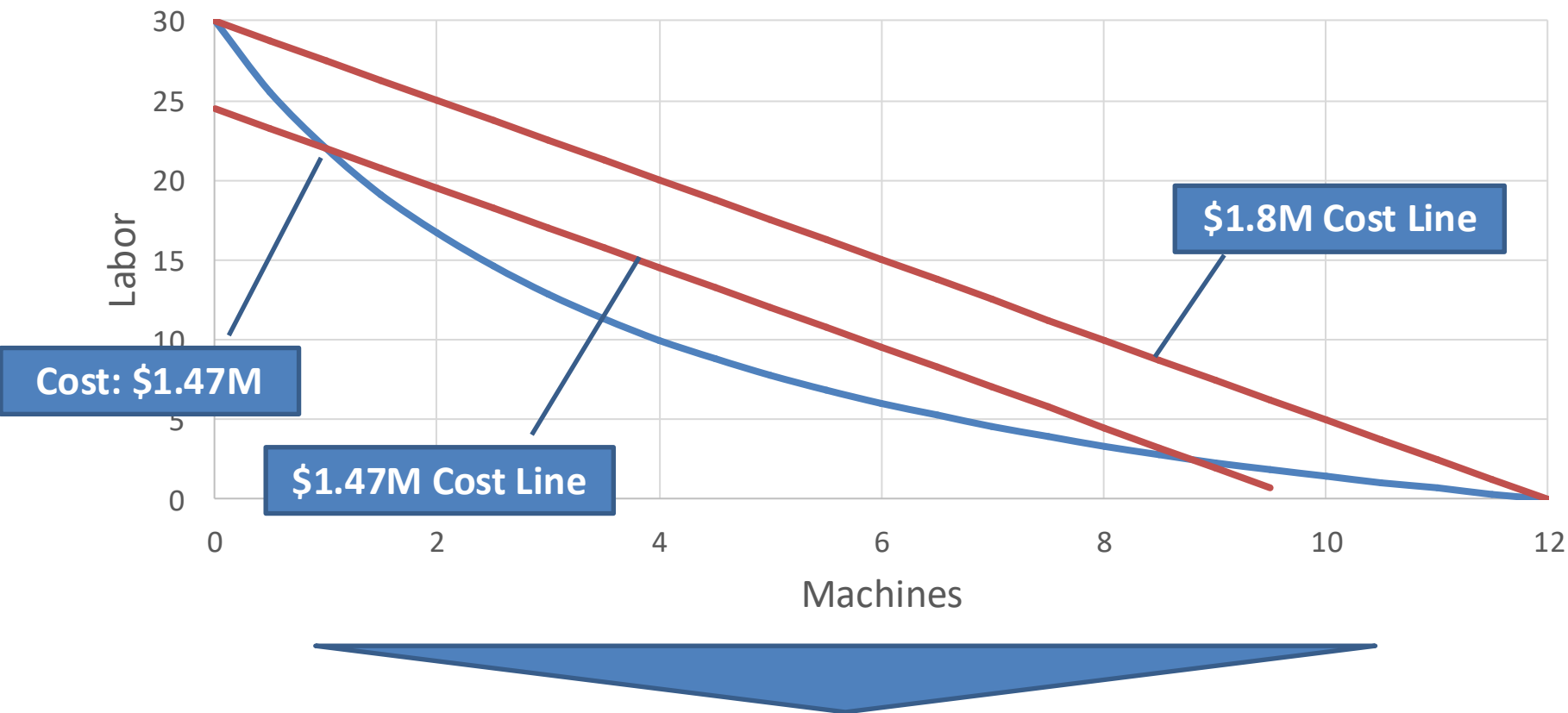
Any point on this line: cost of \$1.8M
Slope of this line = $-p_m/p_l = -2.5$



We can graph the different combinations that get us our output, as well as a line that shows the cost of combinations of inputs.
Slope of the line is the price ratio of the inputs.

Graphically, What Are We Doing?

Production Plans that Achieve 60M Output

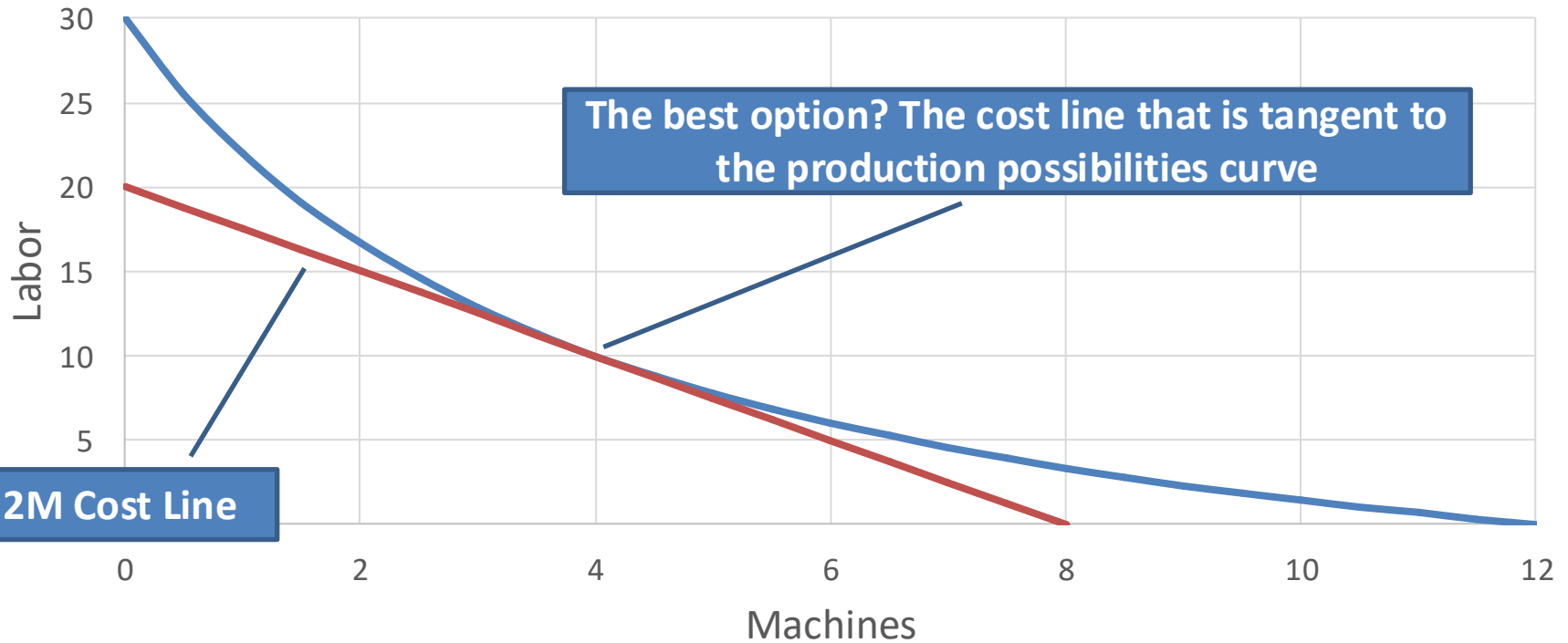


We can draw different cost lines with the same slope (ratio of input prices) that give us different combinations that all have the same cost.

The firm wants to pick the combination with the lowest cost. Where will that be?

Graphically, What Are We Doing?

Production Plans that Achieve 60M Output



The point of tangency of these two curves is the low-cost option to achieve the given level of output.

Graphical Approach is the Same As MPI Approach

We just saw that the optimal point is the tangency point between the two curves. That means their slopes are equal:

Slope of Cost Curve  Slope of Production Curve

$$-\frac{p_m}{p_l} \img alt="double blue equals sign" data-bbox="468 343 515 386"/> -\frac{MP_m}{MP_l}$$

This comes from
implicit differentiation



Rearranging this (and multiplying by -1), we get back to our formula:

$$\frac{MP_l}{p_l} \img alt="double blue equals sign" data-bbox="468 666 515 709"/> \frac{MP_m}{p_m}$$

$$MPI_l \img alt="double blue equals sign" data-bbox="468 770 515 813"/> MPI_m$$

For completeness (none of this math is tested):

Solve for how many machines,
labor as a function of q

- We solved for $q = 60$
- Suppose q is unknown – could solve for how many machines and labor for that unknown q
 - We get functions that describe how many inputs to use to achieve q output at minimum cost.
- Call these functions $m(q)$ and $l(q)$

Construct the firm's cost
function

- Cost is (number of inputs) * (price of inputs)
- So,

$$VC(q) = p_l \cdot l(q) + p_m \cdot m(q)$$

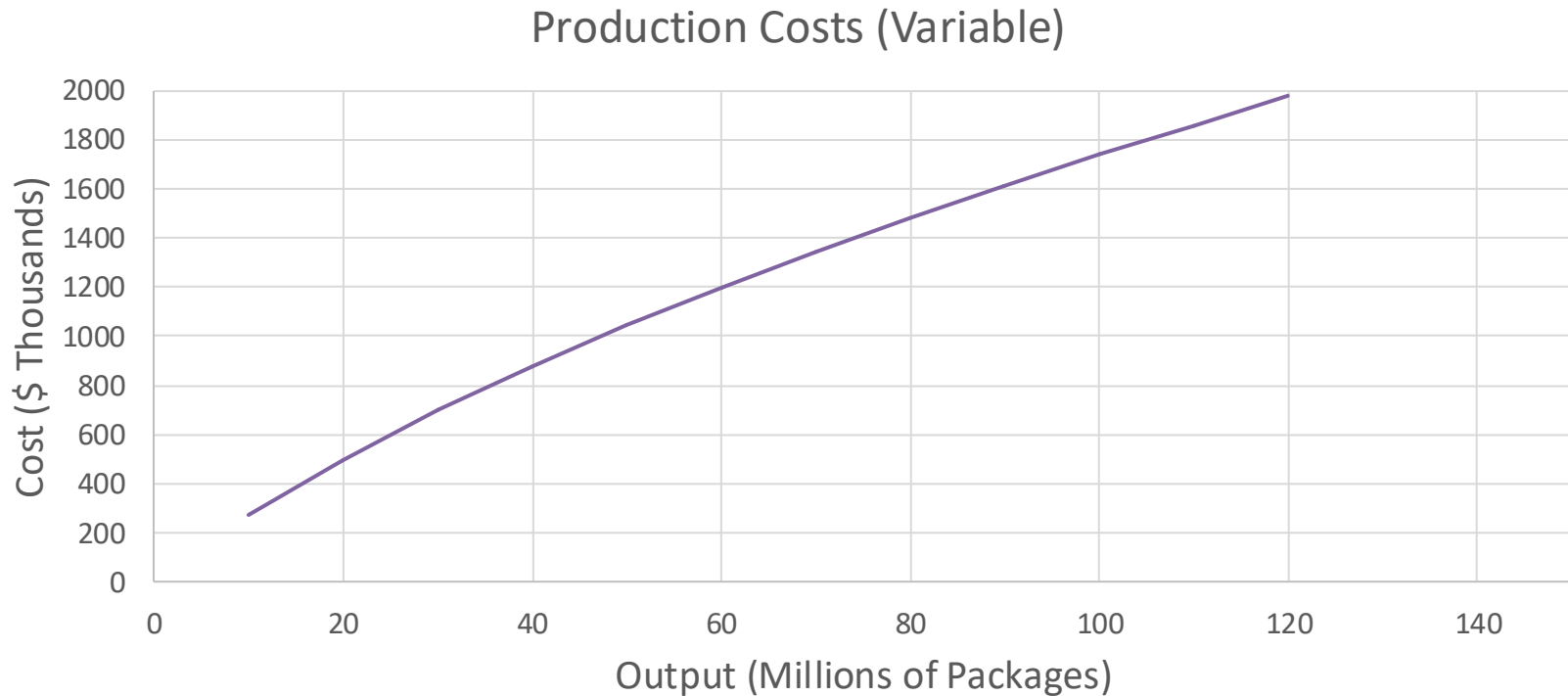
For this example (math is in appendix to these slides), this works out to:

$$VC(q) = 120\sqrt{5}\sqrt{20 + q} - 1200$$

Recall that $MC(q) = \frac{\partial VC(q)}{\partial q}$, so that we have

$$MC(q) = \frac{60\sqrt{5}}{\sqrt{20 + q}}$$

Now we have



We have now solved for the total cost function: for any output level, we now know the lowest possible cost to achieve it.

How would you describe this curve? (Note – it is not a straight line)

Input Prices Can Change...

We solved for the fact that Amazon should use 2.5 workers per robot in their warehouse.

- Note that workers are both substitutes (I can make up for workers with robots) but also complements with robots (workers are more productive with more robots).

What would Amazon do if workers became 10% more expensive?

As an input becomes more expensive...

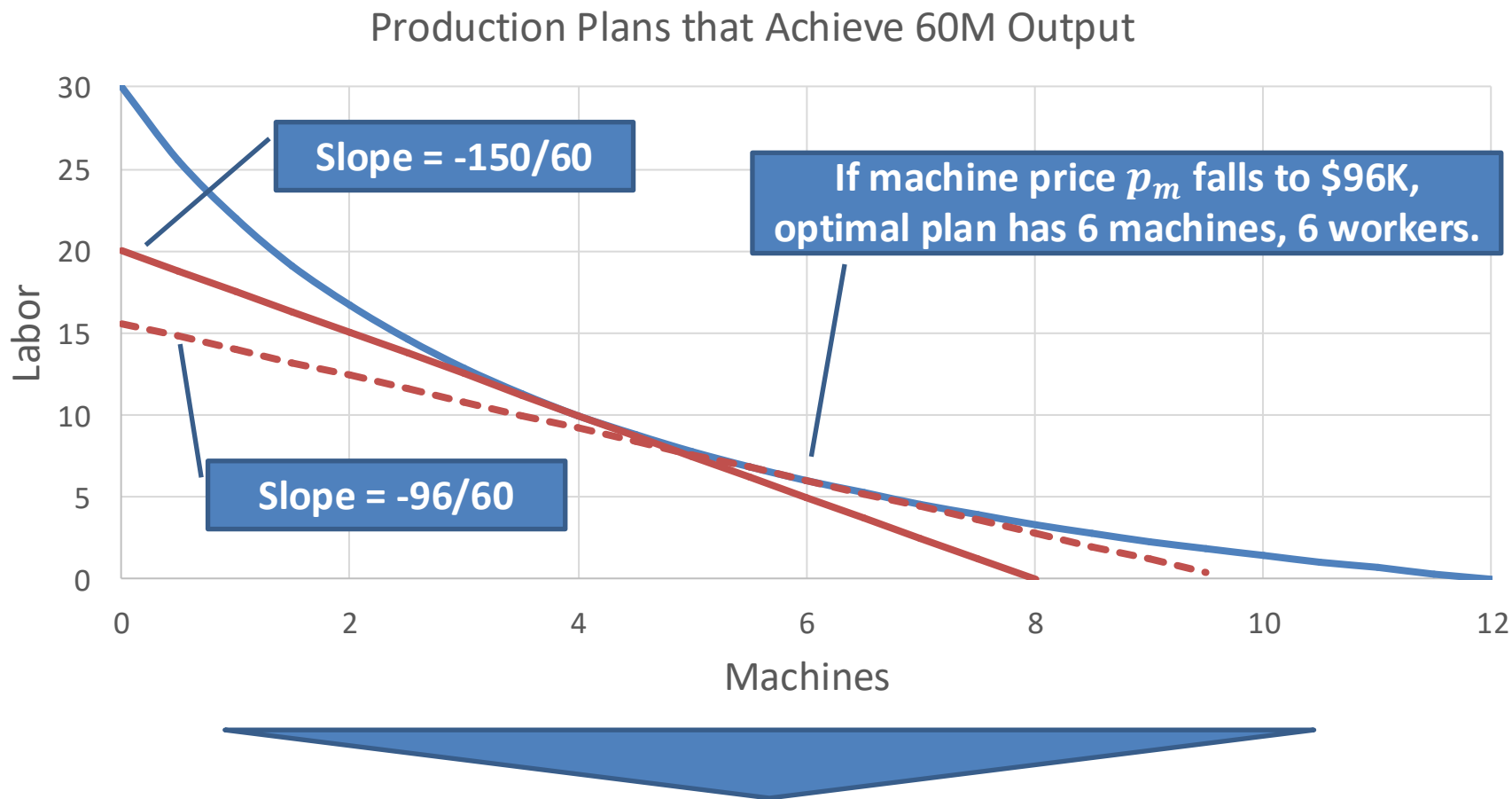
... I substitute *away* from it, toward substitutes.

The complement angle limits how much I substitute away.

Note: this is another elasticity: *elasticity of substitution*.

What does this suggest happens as robots become more *inexpensive*?

Graphically, What Are Input Price Changes?



The slope of the cost curve is the ratio of input prices. So, changing input prices means changing the slope, and finding a different point of tangency.

Suppose $p_m = \$96K$ now. Optimal plan changes to 6 machines, 6 workers.

What happens when input prices change?

Seattle's Minimum Wage Experiment

- What is the effect of an increase in the minimum wage?
 - Seattle: average monthly earnings for some unskilled workers fall when minimum wage goes from \$11 to \$13. (Vigdor et al 2017). Nuanced effects.
 - Most academic research on minimum wage increases to lower levels show positive effects (at lower wage levels).

Short run effects of an input price increase

- Attempt to shift away from that input to substitutes in production.

Long run effects of an input price increase

- Invest in developing new substitutes (e.g. Kiva).



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Amazon Discussion: Capital – Labor tradeoffs (Part 2)

What kinds of costs
are they cutting?

Employees

Rationale?

“But this summer, squat orange robots, called Kiva, began zooming around the shelves instead, picking up goods and carrying them to Rosales at his station. The result? What used to take hours of walking can happen in mere minutes instead.”



In the context of today's lecture, this means that...

$$Q_{new} = f(equip, workers, mgrs, ...) \neq Q_{old} = f(equip, workers, mgrs, ...)$$

New technology has changed the substitutability between people and machines to accomplish the same task.

BUSINESS DAY

The Next Big Leap: The Robots of Tomorrow Are Taking Care of Themselves

By CADE METZ

BERKELEY, Calif. — The robot, perched over a bin filled with objects, from a box of instant small toy shark. This two-armed did not recognize any of this stuff did not matter. It reached into started picking things up, one after another.

“It figures out the best way to object, right from the middle said Jeff Mahler, one of the researchers developing the robot inside a University of California, Berkeley.

For the typical human, that is. For a robot, it is a remarkable something that could drive significant changes inside some of the world’s businesses and further shift the

Today, robots play important manufacturing companies like very specific tasks, like moving placing a particular chip on a stuff, or accomplish more complex centers — where sorting through

BUSINESS DAY

The Robots of Tomorrow Are Taking Care of Themselves

By NICK WINGFIELD

FLORENCE, N.J. — Niss working at the cavernous warehouse in southern New year, stacking plastic bin ottomans. It was not, she stimulating activity. And which often weigh 25 pounds tiring over 10-hour shifts.

Now Ms. Scott, 21, watches replacement — a giant, b mechanical arm — do the

Her new job at Amazon is robots at a time, trouble when necessary and making bins to load. On a recent at end of the arm grabbed conveyor belt and stacking forming neat columns or time Amazon had shown warehouses, to a reporter

“For me, it’s the most mentally challenging thing we have here,” Ms. Scott said of her new job. “It’s not repetitive.”

Perhaps no company embodies the anxieties and hopes around automation better than Amazon. Many people, including President Trump, blame the company for destroying traditional retail jobs by enticing people to shop online. At the same

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BUSINESS

Amazon to Retrain a Third of Its U.S. Workforce

As technology reshapes roles, the online retailer plans to train 100,000 workers in new skills, from machine learning to



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WSJ: Don't Fear Robots

“Robot Apocalypse”

- 1589: Queen refuses patent to inventor of automatic knitting machine, concerned over lost jobs
- 1970s: ATMs were going to put bank tellers out of work. Bank branches employed 30% fewer tellers, but there were 43% more branches.
- 1980s: software makes typesetting easier. 100,000 typesetters lose jobs, but number of jobs for designers increases by 800,000.
- Today: retail jobs are disappearing. Fulfillment jobs are growing.



Automation is fine for employment as long as there is plenty of demand for the good (automation and humans are “complements” more than “substitutes”)

Automation is a problem if the market is “satiated”. Then automation is substitution, not a complement.

What is the situation in retail?
Why?

Are Human Jobs Doomed?



Bilal Farooqui @bilalfarooqui · Jul 17

Our D.C. office building got a security robot. It drowned itself.

We were promised flying cars, instead we got suicidal robots.



4.8K 136K 300K

... not yet.

Summary

Today's Recap

- Types of Costs
- The Production Function
- Cost Minimization
 - Finding “Marginal Product”: **Partial derivative of production function**
 - Equating MPI to find optimal production function: **$MPI = MP / \text{cost}$**
 - Solving for inputs in terms of quantity, and then costs as a function of quantity: **solve $q = f(\text{inputs})$ to get $\text{inputs}(q)$, multiply $\text{inputs}(q)$ by price to get VC**
 - Taking the derivative to get “best” Marginal Cost: **derivative of VC (or TC!)=MC**
- Long-run versus Short-run Costs

Trade

- Theory of Comparative Advantage
- Autarky vs Free Trade
- Trade policy analysis

Step 2 (Cont): Finding required input amounts

Solve for m^* as a function of a general unknown quantity q :

$$q = 10m^* + 1.25m^{*2}$$

$$1.25m^{*2} + 10m^* - q = 0 \quad \leftarrow \text{Rearrange}$$

Math not
tested – just for
completeness!

$$m^* = \frac{-10 \pm \sqrt{100 + 5q}}{2.5} \quad \leftarrow \text{Quadratic formula}$$

$$m^* = 0.4\sqrt{5}\sqrt{20 + q} - 4$$

Now, instead of knowing how much “ q ” I can make with a certain amount of m , I know how much m I need to get q out. I have a function $m^*(q)$.

To find how much l I need, just use the optimal input ratio from step 1:

$$l^* = 2.5m^* = \sqrt{5}\sqrt{20 + q} - 10 \quad \leftarrow \text{Plug in for } m^*(q)$$

$$m^*(q) = 0.4\sqrt{5}\sqrt{20 + q} - 4, \quad l^*(q) = \sqrt{5}\sqrt{20 + q} - 10$$

If I want to produce q output, I should use $m^*(q)$ machines and $l^*(q)$ workers.

Step 3: Multiplying by price to find variable cost

The Concept

- Now we can multiply those input amounts by their “price” or “cost” to find $VC(q)$

Example

If Amazon wants to ship 60m packages...

- Optimal input levels for $q = 60$ are $m^* = 4 \rightarrow l^* = 10$

To find cost, we multiply these input levels by the salaries/rents required

- $VC(60) = \$60K \cdot l^* + \$150K \cdot m^* = \$60K \cdot 10 + \$150K \cdot 4 = \$1.2M$

Math not
tested – just for
completeness

For general q , multiply $m^*(q)$ and $l^*(q)$ by their costs:

$$m^*(q) = 0.4\sqrt{5}\sqrt{20 + q} - 4, \quad l^*(q) = \sqrt{5}\sqrt{20 + q} - 10$$

$$VC(q) = 60(l^*(q)) + 150(m^*(q)) \quad \leftarrow \text{Cost/unit} \cdot \text{optimal units at any production level, } q$$

$$VC(q) = 60(\sqrt{5}\sqrt{20 + q} - 10) + 150(0.4\sqrt{5}\sqrt{20 + q} - 4)$$

$$VC(q) = 120\sqrt{5}\sqrt{20 + q} - 1200$$

$\leftarrow$ Substitute in l^* and m^*

Step 3: Variable Cost to Marginal Cost

Remember, we were calculating the minimal variable cost function in order to get to marginal cost...

$$MC(q) = \frac{\partial VC(q)}{\partial q}$$


- Take derivative to get an optimal Marginal Cost function

$$MC(q) = \frac{\partial VC(q)}{\partial q} = \frac{\partial (120\sqrt{5}\sqrt{20+q} - 1200)}{\partial q} = \frac{60\sqrt{5}}{\sqrt{20+q}}$$

- To get Total Cost function instead, add F (from another source) to VC.
- To find AVC, divide VC by Q.
- To find ATC, divide TC by Q.